

Tutorial Topic: Transport into and out of the cell

Due: 11:59pm on Thursday, October 3, 2024

You will receive no credit for items you complete after the assignment is due. [Grading Policy](#)

Get Ready for This Chapter: Chapter 8 Question 1

In this chapter, you will learn about the organization, structure, and function of cell membranes, including regulation of transport into and out of the cell. Before beginning this chapter, you should be able to explain hydrophilic and hydrophobic properties of molecules (see Concept 3.2, section entitled "Hydrophilic and Hydrophobic Substances"). The following question provides a quick check of your basic knowledge in this area.

Part A

Hydrophobic substances like salad oil are

ANSWER:

- ☐ polar molecules that do not have an affinity for water molecules.
- ☐ polar molecules that have an affinity for water molecules because they contain many hydrogens that can form hydrogen bonds with water.
- ☐ nonpolar molecules that have an affinity for water because they contain many hydrogens that can form hydrogen bonds with water.
- ☐ nonpolar molecules that do not have an affinity for water molecules.

Get Ready for This Chapter: Chapter 8 Question 2

In this chapter, you will learn about the organization, structure, and function of cell membranes, including regulation of transport into and out of the cell. Before beginning this chapter, you should be able to describe the structure of a phospholipid (see Concept 5.3, section entitled "Phospholipids"). The following question provides a quick check of your basic knowledge in this area.

Part A

How do membrane phospholipids interact with water?

ANSWER:

- ☐ The polar heads interact with water; the nonpolar tails do not.
- ☐ Phospholipids do not interact with water because they are lipids, and thus are completely hydrophobic.
- ☐ They have hydrophilic tails that face outward and are exposed to water and hydrophobic heads that face the center of the membrane and are shielded from water.
- ☐ The polar heads have no an affinity for water and the nonpolar tails have an affinity for water.

Chapter 8 Pre-Test Question 1

Part A

Which of the following statements about a typical plasma membrane is correct?

You did not open hints for this part.

ANSWER:

- ☐ The hydrophilic interior of the membrane is composed primarily of the fatty acid tails of the phospholipids.
- ☐ The two sides of the plasma membrane have different lipid and protein composition.
- ☐ Carbohydrates on the membrane surface are important in determining the overall bilayer structure.
- ☐ The plasma membrane is a covalently linked network of phospholipids and proteins that controls the movement of solutes into and out of a cell.
- ☐ Phospholipids are the primary component that determines which solutes can cross the plasma membrane.

Chapter 8 Pre-Test Question 2

Part A

Which of the following *best* describes the structure of a biological membrane?

You did not open hints for this part.

ANSWER:

- ☐ two layers of phospholipids (with opposite orientations of the phospholipids in each layer), with each layer covered on the outside with proteins
- ☐ two layers of phospholipids, with proteins either crossing the layers or on the surface of the layers
- ☐ two layers of phospholipids, with proteins embedded between the two layers
- ☐ a mixture of covalently linked phospholipids and proteins that determines which solutes can cross the membrane and which cannot
- ☐ a fluid structure in which phospholipids and proteins move freely between sides of the membrane

Chapter 8 Question 2

Part A

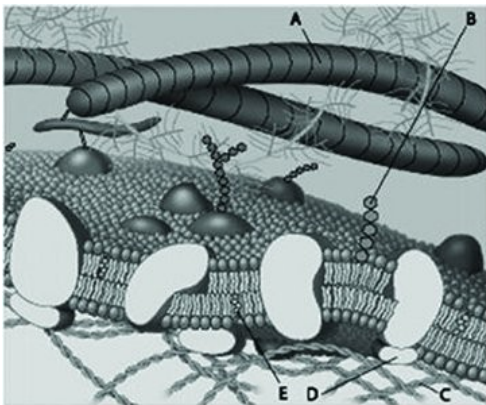
A phospholipid bilayer with equal amounts of saturated and unsaturated fatty acids displays a specific permeability to glucose. What effect will increasing the proportion of unsaturated fatty acids in the bilayer have on the membrane's permeability to glucose?

ANSWER:

- ☐ Permeability to glucose will increase.
- ☐ Permeability to glucose will decrease.
- ☐ Permeability to glucose will stay the same.
- ☐ Permeability will decrease initially then increase as the bilayer fills with glucose.

Chapter 8 Question 10

Part A



Which component in the figure above is hydrophilic?

ANSWER:

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Chapter 8 Pre-Test Question 4

Part A

Which of the following is *least* likely to be important in holding the components of a biological membrane together?

You did not open hints for this part.

ANSWER:

- ☐ hydrophobic interactions between the phospholipid tails and the surface of integral membrane proteins buried in the membrane
- ☐ polar interactions among the phospholipid head groups on the same surface of the membrane
- ☐ hydrophobic interactions among the fatty acid tails of phospholipids on the same side of the membrane
- ☐ hydrophobic interactions among the fatty acid tails of phospholipids on opposite sides of the membrane
- ☐ covalent interactions between the phospholipid and protein components of the membrane

Chapter 8 Question 17

Part A

Which of the following statements best describes how unsaturated fatty acids increase membrane fluidity at lower temperatures?

ANSWER:

- ☐ The double bonds block interaction among the hydrophilic head groups of the lipids.
- ☐ The double bonds form kinks in the fatty acid tails, preventing adjacent lipids from packing tightly.
- ☐ Unsaturated fatty acids are more nonpolar than saturated fatty acids.
- ☐ Unsaturated fatty acids have a higher cholesterol content, which prevents adjacent lipids from packing tightly.

Chapter 8 Question 1

Part A

Which of the following statements best describes a characteristic feature of integral membrane proteins?

ANSWER:

- ☐ They are amphipathic, with at least one hydrophobic region.
- ☐ They are hydrophilic.
- ☐ They are localized to the interior surface of the membrane.
- ☐ They are hydrophobic.

Chapter 8 Question 6

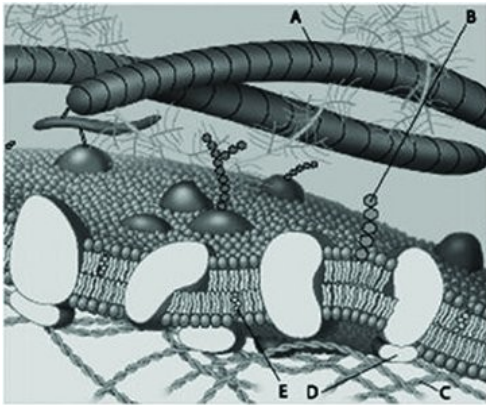
Part A

An animal cell that lacks carbohydrates on the external surface of its plasma membrane would likely be impaired in which of the following functions?

ANSWER:

- ☐ transporting ions against an electrochemical gradient
- ☐ attaching the plasma membrane to the cytoskeleton
- ☐ cell-cell recognition
- ☐ establishing a diffusion barrier to charged molecules

Chapter 8 Question 14

Part A

Which component in the accompanying figure plays a critical role in cell-cell recognition?

ANSWER:

- ☐ A
- ☐ B
- ☐ C
- ☐ E

Chapter 8 Question 5

Part A

Which of the following statements best describes the structure of a membrane based on the fluid mosaic model?

ANSWER:

- ☐ The membrane is composed of a single layer of fluid phospholipids between two layers of hydrophilic proteins.
- ☐ The membrane is composed of a mosaic of fluid polysaccharides and amphipathic proteins.
- ☐ The membrane is composed of a fluid bilayer of phospholipids between two layers of hydrophilic proteins.
- ☐ The membrane is composed of a fluid bilayer of phospholipids with embedded amphipathic proteins.

Chapter 8 Question 7

Part A

Which of the following types of molecules are most likely to be excluded from the hydrophobic portion of the phospholipid bilayer?

ANSWER:

- ☐ integral membrane proteins
- ☐ peripheral membrane proteins
- ☐ cholesterol
- ☐ transmembrane proteins

Chapter 8 Question 2

Part A

According to the fluid mosaic model of membrane structure, proteins of the membrane are mostly

ANSWER:

- ☐ spread in a continuous layer over the inner and outer surfaces of the membrane.
- ☐ embedded in a lipid bilayer.
- ☐ randomly oriented in the membrane, with no fixed inside-outside polarity.
- ☐ confined to the hydrophobic interior of the membrane.

Chapter 8 Question 9

Part A

Which of the following statements best explains the reason that lipids and proteins are free to move laterally in membranes?

ANSWER:

- ☐ There are only weak hydrophobic interactions in the interior of the membrane.
- ☐ The interior of the membrane is filled with liquid water.
- ☐ Hydrophilic portions of the lipids are in the interior of the membrane.
- ☐ Lipids and proteins repulse each other in the membrane.

Chapter 8 Question 4

Part A

Which of the following statements best explains how the membranes in winter wheat are able to remain fluid in extremely cold temperatures?

ANSWER:

- ☐ The membranes contain a reduced proportion of hydrophobic proteins.
- ☐ The membranes contain an increased proportion of unsaturated phospholipids.
- ☐ The membranes contain a reduced proportion of cholesterol molecules.
- ☐ The membranes contain an increased proportion of glycolipids.

Chapter 8 Question 15

Part A

Cell membranes have distinct inside and outside faces. Which of the following statements is the most plausible explanation for the asymmetrical nature of the two layers of the membrane?

ANSWER:

- ☐ Cell membranes communicate signals between different cell types.
- ☐ Proteins only function on the cytoplasmic side of the cell membrane.
- ☐ Cell membranes form a border between adjacent cells in tightly packed tissues such as epithelium.
- ☐ The two layers of a cell membrane face different environments and carry out different functions.

Chapter 8 Question 3

Part A

Which of the following factors would tend to increase membrane fluidity?

ANSWER:

- ☐ a greater proportion of saturated phospholipids
- ☐ a greater proportion of unsaturated phospholipids
- ☐ a lower temperature
- ☐ a relatively high protein content in the membrane

Chapter 8 Question 8

Part A

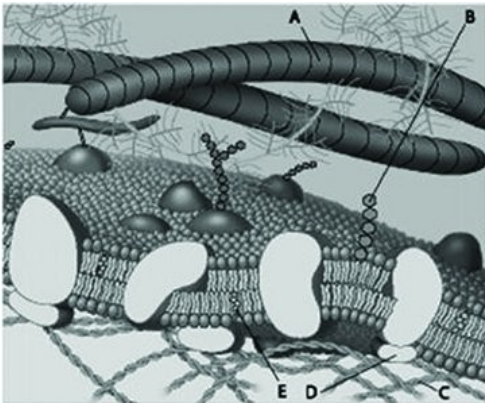
Which of the following types of molecules are most likely to lack hydrophilic domains?

ANSWER:

- ☐ transmembrane proteins
- ☐ cholesterol
- ☐ ion transport proteins
- ☐ peripheral membrane proteins

Chapter 8 Question 11

Part A



Which component in the figure above represents cholesterol?

ANSWER:

- ☐ E
- ☐ D
- ☐ C
- ☐ B

Chapter 8 Question 1

Part A

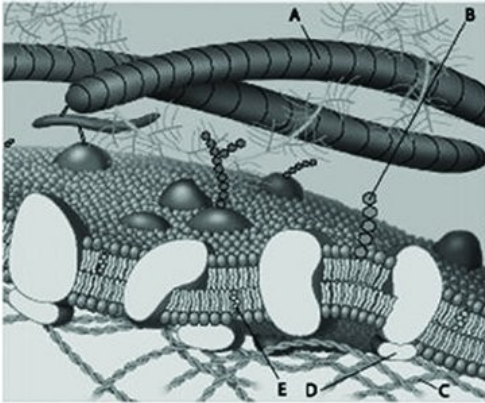
In what way do the membranes of a eukaryotic cell vary?

ANSWER:

- ☐ Only certain membranes of the cell are selectively permeable.
- ☐ Only certain membranes are constructed from amphipathic molecules.
- ☐ Phospholipids are found only in certain membranes.
- ☐ Certain proteins are unique to each membrane.

Chapter 8 Question 13

Part A



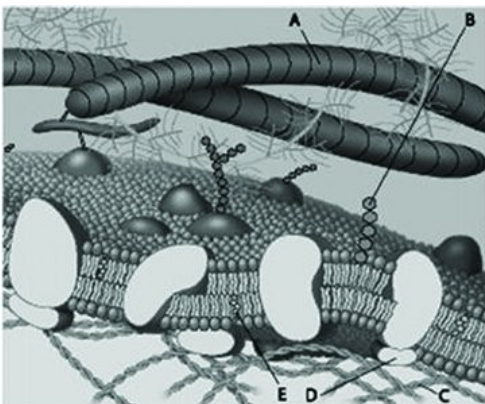
Which component in the figure above represents a microfilament of the cytoskeleton?

ANSWER:

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Chapter 8 Question 12

Part A



Which component in the figure above represents a protein fiber of the extracellular matrix?

ANSWER:

- ☐ A
- ☐ B
- ☐ C
- ☐ E

Chapter 8 Pre-Test Question 3

Part A

The permeability of a biological membrane to a specific polar solute may depend on which of the following?

You did not open hints for this part.

ANSWER:

- ☐ the types of polysaccharides present in the membrane
- ☐ the presence of unsaturated fatty acids in the membrane
- ☐ the amount of cholesterol in the membrane
- ☐ the phospholipid composition of the membrane
- ☐ the types of transport proteins in the membrane

Chapter 8 Question 23

Part A

A new drug designed for the treatment of lung cancer must specifically enter the cytoplasm of lung cells while not entering the cells of other tissues. Which of the following characteristics would likely enhance the specificity of this drug?

ANSWER:

- ☐ the specificity of the drug molecule for binding to the extracellular matrix of lung cells
- ☐ the relative hydrophobicity of the drug molecule
- ☐ the similarity of the drug molecule to other molecules normally transported lung cells
- ☐ the phospholipid composition of lung cell membranes

Chapter 8 Question 20

Part A

Which of the following statements best describes a characteristic feature of a carrier protein in a plasma membrane?

ANSWER:

- ☐ It exhibits specificity for a particular type of molecule.
- ☐ It works against a concentration gradient.
- ☐ It requires the expenditure of cellular energy to function.
- ☐ It has no hydrophobic regions.

Chapter 8 Question 22

Part A

A defect in which of the following molecules would dramatically reduce the rate of diffusion of water across cell membranes?

ANSWER:

- ☐ gated ion channels
- ☐ aquaporins
- ☐ sodium-potassium pump
- ☐ ATP

Chapter 8 Question 21

Part A

Which of the following molecules would likely diffuse through the lipid bilayer of a plasma membrane most rapidly?

ANSWER:

- ☐ O₂
- ☐ an amino acid
- ☐ sucrose
- ☐ Na⁺

Chapter 8 Question 19

Part A

Which of the following types of molecules are *least* likely to diffuse through the phospholipid bilayer of a cell membrane?

ANSWER:

- ☐ small hydrophobic molecules
- ☐ carbon dioxide
- ☐ large hydrophobic molecules
- ☐ small ions

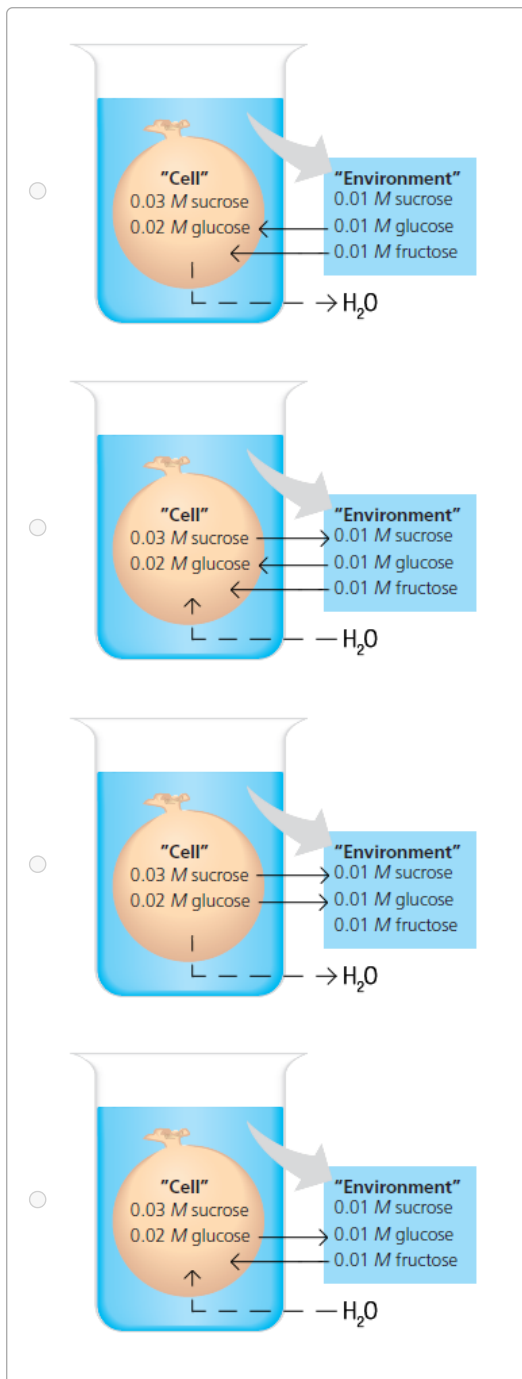
Chapter 8 Question 6

An artificial "cell" consisting of an aqueous solution enclosed in a selectively permeable membrane is immersed in a beaker containing a different solution, the "environment," as shown below. The membrane is permeable to water and to the simple sugars glucose and fructose but impermeable to the disaccharide sucrose.

Part A

Identify the correct net movement of solutes into and/or out of the cell (labeled with solid arrows) and the net osmosis (labeled with dashed arrows).

ANSWER:



Part B Complete previous part(s)

Instructors: [View all hidden parts](#)

Part C Complete previous part(s)

Instructors: [View all hidden parts](#)

Part D Complete previous part(s)

Instructors: [View all hidden parts](#)

Chapter 8 Question 11

Part A

In the supermarket, lettuce and other produce are often sprayed with water. Why does this make vegetables crisp?

Match the terms in the left column to the appropriate blanks in the sentences on the right. Not all terms will be used.

ANSWER:

Reset Help

hypertonic

into

hypotonic

mineral ions

water

out of

Vegetables are crisp largely because of [] trapped within them. The cells in lettuce are [] to freshwater. The plant cells are placed in a [] solution when vegetables are sprayed with water. In this case, water will move [] the plant cells, and the vegetables will be crisp.

Chapter 8 Question 41

Part A

Three lab groups carried out an experiment to identify the concentration of sucrose in six solutions. Each unknown solution (1-6) contained one of the following sucrose concentrations: 0.0 M, 0.2 M, 0.4 M, 0.6 M, 0.8 M, and -1.0 M. Cubes of sweet potato (1 cm^3) were soaked for 24 hours in each solution and weighed to determine the change in mass. Each data entry represents the average of three sample replicates expressed as percent change in mass following a 24-hour soak in the unknown solutions. From the data given, which statement below most accurately describes something that has occurred in response to a particular unknown solution?

Unknown	Percent Change in Mass	Percent Change in Mass	Percent Change in Mass
	Group 1	Group 2	Group 3
1	6.6	7.8	7.5
2	3.1	3.7	2.9
3	-2.7	-3.5	-2.5
4	0.7	0.5	1.1
5	-11.6	-12.3	-12.6
6	-5.2	-6.2	-4.9

ANSWER:

- ☐ Unknown solution 5 contains the highest concentration of sucrose, and the change in mass is due to the active transport of sucrose out of the cell.
- ☐ Unknown solution 3 represents a sucrose concentration slightly higher than the molarity of sweet potato cells, thus water is transported out of the cells.
- ☐ Passive transport of sucrose out of the potato cells explains the change in mass observed for unknown solution 6.
- ☐ Osmosis of water molecules from unknown solution 2 likely caused the increase in mass observed.

Chapter 8 Pre-Test Question 5

Part A

Which of the following statements about osmosis is correct?

You did not open hints for this part.

ANSWER:

- ☐ Osmosis is the diffusion of water from a region of lower water concentration to a region of higher water concentration.
- ☐ If a solution outside the cell is hypertonic compared to the cytoplasm, water will move into the cell by osmosis.
- ☐ The presence of aquaporins (proteins that form water channels in the membrane) speeds up the process of osmosis.
- ☐ If a cell is placed in an isotonic solution, more water will enter the cell than leaves the cell.
- ☐ Osmotic movement of water into a cell would likely occur if the cell accumulates water from its environment.

Chapter 8 Question 40

Part A

Three lab groups carried out an experiment to identify the concentration of sucrose in six solutions. Each unknown solution (1-6) contained one of the following sucrose concentrations: 0.0 M, 0.2 M, 0.4 M, 0.6 M, 0.8 M, and -1.0 M. Cubes of sweet potato (1 cm³) were soaked for 24 hours in each solution and weighed to determine the change in mass. Each data entry represents the average of three sample replicates expressed as percent change in mass following a 24-hour soak in the unknown solutions. From the data provided, which unknown solution has a molarity that most closely reflects the molarity of sweet potato cells?

Unknown	Percent Change in Mass	Percent Change in Mass	Percent Change in Mass
	Group 1	Group 2	Group 3
1	6.6	7.8	7.5
2	3.1	3.7	2.9
3	-2.7	-3.5	-2.5
4	0.7	0.5	1.1
5	-11.6	-12.3	-12.6
6	-5.2	-6.2	-4.9

ANSWER:

☐ 4

☐ 3

☐ 2

☐ 5

Chapter 8 Question 42

Part A

Three lab groups carried out an experiment to identify the concentration of sucrose in six solutions. Each unknown solution (1-6) contained one of the following sucrose concentrations: 0.0 M, 0.2 M, 0.4 M, 0.6 M, 0.8 M, and 1.0 M. Cubes of sweet potato (1 cm³) were soaked for 24 hours in each solution and weighed to determine the change in mass. Each data entry represents the average of three sample replicates expressed as percent change in mass following a 24-hour soak in the unknown solutions.

Unknown	Percent Change in Mass	Percent Change in Mass	Percent Change in Mass
	Group 1	Group 2	Group 3
1	6.6	7.8	7.5
2	3.1	3.7	2.9
3	-2.7	-3.5	-2.5
4	0.7	0.5	1.1
5	-11.6	-12.3	-12.6
6	-5.2	-6.2	-4.9

Which unknown solution contains the highest concentration of sucrose?

ANSWER:

☐ 1

☐ 4

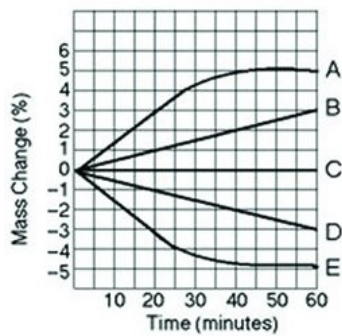
☐ 3

☐ 5

Chapter 8 Question 33

Part A

Five dialysis bags, constructed of a type of membrane that is permeable to water and impermeable to sucrose, were filled with various concentrations of sucrose and then placed in separate beakers containing an initial concentration of 0.6 M sucrose solution. At 10-minute intervals, the bags were weighed, and the percent change in mass of each bag was graphed.



Which line or lines in the graph represent(s) bags that contain a solution that is hypertonic to the solution in the beaker at 50 minutes?

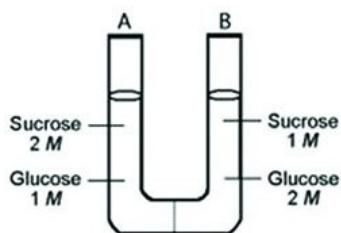
ANSWER:

- ☐ D
- ☐ B
- ☐ D and E
- ☐ A and B

Chapter 8 Question 28

Part A

The solutions in the two arms of a U-tube are separated by a membrane that is permeable to water and glucose but not to sucrose. Side A is half-filled with a solution of 2 M sucrose and 1 M glucose. Side B is half-filled with 1 M sucrose and 2 M glucose. Initially, the liquid levels on both sides are equal.



Which of the following statements best describes the relationship between the solutions on both sides of the U-tube?

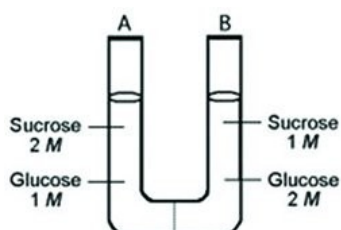
ANSWER:

- ☐ side A is more turgid than side B
- ☐ side A is hypotonic to side B
- ☐ side A is isotonic to side B
- ☐ side A is hypertonic to side B

Chapter 8 Question 30

Part A

The solutions in the two arms of a U-tube are separated by a membrane that is permeable to water and glucose but not to sucrose. Side A is half-filled with a solution of 2 M sucrose and 1 M glucose. Side B is half-filled with 1 M sucrose and 2 M glucose. Initially, the liquid levels on both sides are equal.



When the system illustrated above reaches equilibrium, the sugar concentrations on both sides of the U-tube will be _____.

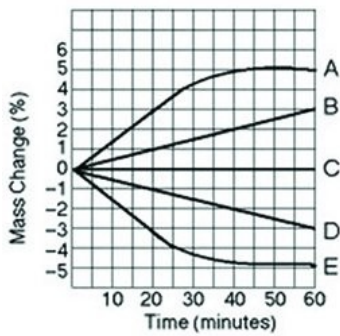
ANSWER:

- ☐ 1.5 M sucrose, 1.5 M glucose
- ☐ 2 M sucrose, 1 M glucose
- ☐ 1 M sucrose, 1 M glucose
- ☐ 1 M sucrose, 2 M glucose

Chapter 8 Question 32

Part A

Five dialysis bags, constructed of a type of membrane that is permeable to water and impermeable to sucrose, were filled with various concentrations of sucrose and then placed in separate beakers containing an initial concentration of 0.6 M sucrose solution. At 10-minute intervals, the bags were weighed, and the percent change in mass of each bag was graphed.



Which line in the graph represents the bag with the highest initial concentration of sucrose?

ANSWER:

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Chapter 8 Question 34

Part A

Celery stalks that are immersed in fresh water for several hours become stiff. Which of the following statements best explains the observation that similar stalks left in a 0.15 M salt solution become limp?

ANSWER:

- ☐ The fresh water and the salt solution are both hypertonic to the cells of the celery stalks.
- ☐ The fresh water is hypotonic and the salt solution is hypertonic to the cells of the celery stalks
- ☐ The fresh water is isotonic and the salt solution is hypertonic to the cells of the celery stalks
- ☐ The fresh water is hypertonic and the salt solution is hypotonic to the cells of the celery stalks

Chapter 8 Question 35

Part A

What will happen to a red blood cell (RBC), which has an internal ion content of about 0.9%, if it is placed into a beaker of pure water?

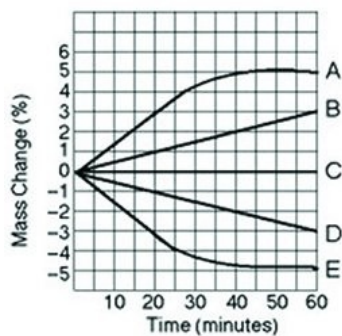
ANSWER:

- ☐ The cell will remain the same size because the solution outside the cell is isotonic.
- ☐ The cell will shrink because the water in the beaker is hypotonic relative to the cytoplasm of the RBC.
- ☐ The cell will shrink because the water in the beaker is hypertonic relative to the cytoplasm of the RBC.
- ☐ The cell will swell because the water in the beaker is hypotonic relative to the cytoplasm of the RBC.

Chapter 8 Question 31

Part A

Five dialysis bags, constructed of a type of membrane that is permeable to water and impermeable to sucrose, were filled with various concentrations of sucrose and then placed in separate beakers containing an initial concentration of 0.6 M sucrose solution. At 10-minute intervals, the bags were weighed, and the percent change in mass of each bag was graphed.



Which line in the graph represents the bag that contained a solution isotonic to the 0.6 M solution at the beginning of the experiment?

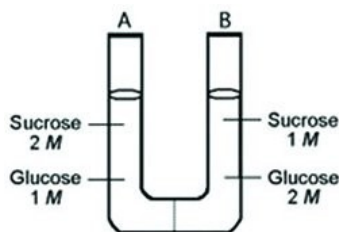
ANSWER:

- ☐ A
- ☐ B
- ☐ C
- ☐ D

Chapter 8 Question 29

Part A

The solutions in the two arms of a U-tube are separated by a membrane that is permeable to water and glucose but not to sucrose. Side A is half-filled with a solution of 2 M sucrose and 1 M glucose. Side B is half-filled with 1 M sucrose and 2 M glucose. Initially, the liquid levels on both sides are equal.



Which of the following is the most likely result when the system has reached equilibrium?

ANSWER:

- ☐ The water level will be higher in side B than in side A.
- ☐ The concentration of sucrose on side A will be greater than the concentration of sucrose on side B.
- ☐ The water levels will be unchanged.
- ☐ The water level will be higher in side A than in side B.

Chapter 8 Question 36

Part A

Which of the following statements best describes the normal tonicity conditions for typical plant and animal cells in their native environments?

ANSWER:

- ☐ The animal cell is in an isotonic solution, and the plant cell is in a hypertonic solution.
- ☐ The animal cell is in a hypotonic solution, and the plant cell is in an isotonic solution.
- ☐ The animal cell is in an isotonic solution, and the plant cell is in a hypotonic solution.
- ☐ The animal cell is in a hypertonic solution, and the plant cell is in an isotonic solution.

Chapter 8 Question 37

Part A

In which of the following circumstances would the cell or organism have the greatest need for osmoregulation?

ANSWER:

- ☐ an animal connective tissue cell bathed in isotonic body fluid
- ☐ a salmon moving from a river into an ocean
- ☐ a plant being grown hydroponically in a watery mixture of designated nutrients
- ☐ a red blood cell surrounded by plasma

Chapter 8 Question 27

Part A

Which of the following statements best describes a characteristic feature of *osmosis*?

ANSWER:

- ☐ The process of osmosis requires energy from ATP hydrolysis.
- ☐ In osmosis, water moves across a membrane from areas of lower solute concentration to areas of higher solute concentration.
- ☐ Osmosis only takes place in red blood cells.
- ☐ In osmosis, solutes move across a membrane from areas of lower water concentration to areas of higher water concentration.

Chapter 8 Question 38

Part A

Which of the following results is most likely when a plant cell is submerged in a hypertonic solution?

ANSWER:

- ☐ The cell will become flaccid.
- ☐ The cell will burst.
- ☐ The cell will become turgid.
- ☐ Plasmolysis will shrink the interior of the cell.

Chapter 8 Question 26

Part A

When the interior of a cell is in equilibrium with its environment, which of the following statements best describes the relative movement of molecules that can diffuse through the plasma membrane?

ANSWER:

- ☐ There is directed movement of molecules into or out of the cell.
- ☐ All movement of molecules across the plasma membrane occurs by active transport.
- ☐ There is random movement of molecules into and out of the cell.
- ☐ There is no movement of molecules into or out of the cell.

Chapter 8 Question 24

Part A

Which of the following statements best describes a characteristic feature of diffusion?

ANSWER:

- ☐ It is very rapid over long distances.
- ☐ It is a passive process in which molecules move from a region of higher concentration to a region of lower concentration.
- ☐ It is an active process in which molecules move from a region of lower concentration to a region of higher concentration.
- ☐ It requires an expenditure of energy by the cell.

Chapter 8 Question 25

Part A

Which of the following cellular processes includes all of the others?

ANSWER:

- ☐ transport of an ion down its electrochemical gradient
- ☐ facilitated diffusion
- ☐ passive transport
- ☐ osmosis

Chapter 8 Question 8

SCIENTIFIC INQUIRY

An experiment is designed to study the mechanism of sucrose uptake by plant cells. Cells are immersed in a sucrose solution, and the pH of the solution is monitored. Samples of the cells are taken at intervals, and their sucrose concentration is measured. The pH is observed to decrease until it reaches a steady, slightly acidic level, and then sucrose uptake begins.

Part A

Evaluate these results.

Rank the steps from first to last. To rank items as equivalent, overlap them.

ANSWER:

Reset Help

H⁺ moves down its electrochemical gradient into the cell.

The gradient becomes sufficient.

A proton pump, using ATP, pumps protons outside.

A steady proton gradient with a higher concentration of H⁺ out of the cell is maintained by the proton pump.

pH decreases.

Sucrose moves into the cell.

First step				Last step

Part B Complete previous part(s)

Instructors: [View all hidden parts](#)

Part C Complete previous part(s)

Instructors: [View all hidden parts](#)

Chapter 8 Question 10

A human pancreatic cell obtains O₂—and necessary molecules such as glucose, amino acids, and cholesterol—from its environment, and it releases CO₂ as a waste product. In response to hormonal signals, the cell secretes digestive enzymes. It also regulates its ion concentrations by exchange with its environment.

Part A

Based on what you have learned about the structure and function of cellular membranes, how does such a cell accomplish these interactions with its environment?

Match the terms in the left column to the appropriate blanks in the sentences on the right. Not all terms will be used.

ANSWER:

Reset Help

proton pump

Ca²⁺

cholesterol

polar molecules

sodium-potassium pump

exocytosis

ions

K⁺

nonpolar molecules

endocytosis

hormones

glucose

diffuse through the lipid bilayer.

require transport proteins.

enters by receptor-mediated .

bind to receptors and trigger the secretion of enzymes by .

move through specific channels, which may be gated.

The actively transports Na⁺ and .

Chapter 8 Question 9

Part A

Extensive irrigation in arid regions causes salts to accumulate in the soil. (When water evaporates, salts that were dissolved in the water are left behind in the soil.) Based on what you learned about water balance in plant cells, explain why increased soil salinity (saltiness) might be harmful to crops.

Match the terms in the left column to the appropriate blanks in the sentences on the right. Not all terms will be used.

ANSWER:

[Reset](#) [Help](#)

water

become flaccid

reduction

increase

hypertonic

hypotonic

mineral ions

undergo plasmolysis

Plant cells are turgid in a environment, where the uptake of is eventually balanced by the wall pushing back on the cell.

A high salt concentration in soil solution is a environment for the epidermal cells of the plant roots. In this case, the plant cells . Thus, salt causes a(n) in the rate and amount of water that the plant roots can take up from the soil.

Chapter 8 Question 43

Part A

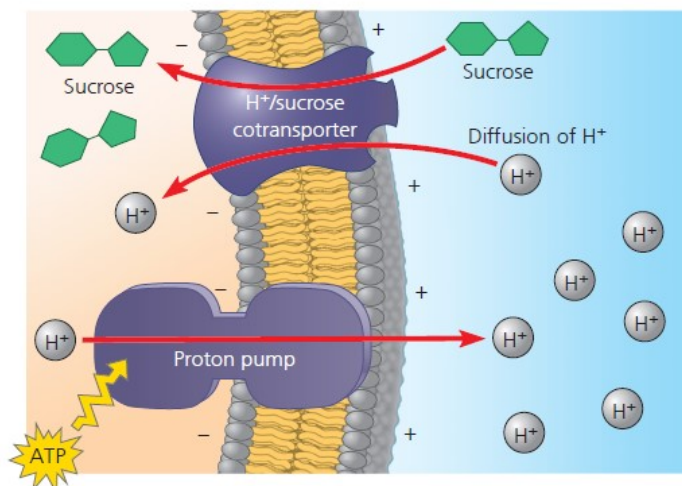
Which of the following statements best describes characteristic activities of a sodium-potassium pump?

ANSWER:

- ☐ The pump moves three sodium ions out of a cell and two potassium ions into a cell using energy from ATP hydrolysis.
- ☐ The pump moves three sodium ions out of a cell and two potassium ions into a cell and generates an ATP in each cycle.
- ☐ The pump moves three potassium ions out of a cell and two sodium ions into a cell while producing ATP for each cycle.
- ☐ The pump moves three potassium ions out of a cell and two sodium ions into a cell using energy from ATP hydrolysis.

Chapter 8 Question 5

Part A



Based on the figure above, which of these experimental treatments would increase the rate of sucrose transport into a plant cell?

ANSWER:

- ☐ decreasing cytoplasmic pH
- ☐ decreasing extracellular sucrose concentration
- ☐ adding a substance that makes the membrane more permeable to hydrogen ions
- ☐ decreasing extracellular pH

Chapter 8 Question 51

Part A

Which of the following statements best describes some aspect of the function of a protein that cotransports glucose and sodium ions into the intestinal cells of an animal?

ANSWER:

- ☐ Sodium ions can move down their electrochemical gradient through the cotransporter whether or not glucose is present outside the cell.
- ☐ Sodium and glucose compete for the same binding site in the cotransporter.
- ☐ Glucose entering the cell down its concentration gradient provides energy for uptake of sodium ions against the electrochemical gradient.
- ☐ A substance that blocks sodium ions from binding to the cotransport protein will also block the transport of glucose.

Chapter 8 Question 52

Part A

Which of the following statements best explains the observation that proton pumps are used by organisms in every domain of life?

ANSWER:

- ☐ The high concentration of protons in the ancient atmosphere must have necessitated a pump mechanism.
- ☐ Cells of each domain evolved proton pumps independently when oceans became more acidic.
- ☐ Proton gradients across a membrane were used by cells that were the common ancestor of all three domains of life.
- ☐ Proton pumps are necessary to all cell membranes.

Chapter 8 Question 50

Part A

In some cells, there are many different ion electrochemical gradients across the plasma membrane even though there are usually only one or two types of proton pumps present in the membrane. The gradients of the other ions are most likely established by which of the following?

ANSWER:

- ☐ cotransport proteins
- ☐ passive diffusion across the plasma membrane
- ☐ pores in the plasma membrane
- ☐ ion channels

Chapter 8 Question 49

Part A

The phosphate transport system in bacteria imports phosphate into the cell even when the concentration of phosphate outside the cell is much lower than the cytoplasmic phosphate concentration. Because phosphate import depends on a pH gradient across the membrane, phosphate transport is most likely an example of which of the following transport processes?

ANSWER:

- ☐ passive diffusion
- ☐ cotransport
- ☐ facilitated diffusion
- ☐ active transport

Chapter 8 Question 44

Part A

Which of the following statements best explains why the sodium-potassium pump is an electrogenic pump?

ANSWER:

- ☐ It transports equal quantities of Na^+ and K^+ across the membrane in opposite directions.
- ☐ It generates voltage across the membrane.
- ☐ It is used to drive the transport of glucose against a concentration gradient.
- ☐ It decreases the voltage difference across the membrane.

Chapter 8 Question 55

Part A

Which of the following statements most accurately compares primary factors responsible for simple diffusion versus active transport?

ANSWER:

- ☐ Simple diffusion is driven by ATP hydrolysis, while active transport is driven by an electrochemical gradient.
- ☐ Simple diffusion is driven by a concentration gradient, while active transport is driven by ADP.
- ☐ Simple diffusion is driven by a concentration gradient, while active transport is driven by ATP hydrolysis.
- ☐ Simple diffusion is driven transmembrane pumps, while active transport is driven by ATP hydrolysis.

Chapter 8 Question 45

Part A

Which of the following membrane activities requires energy from ATP hydrolysis?

ANSWER:

- ☐ facilitated diffusion of chloride ions across the membrane through a chloride channel
- ☐ movement of carbon dioxide out of a paramecium
- ☐ movement of Na^+ ions from a lower concentration in a mammalian cell to a higher concentration in the extracellular fluid
- ☐ movement of glucose molecules into a bacterial cell from a medium containing a higher concentration of glucose than inside the cell

Chapter 8 Question 47

Part A

Diffusion of ions across membranes through specific ion channels is driven by _____.

ANSWER:

- ☐ ion electrochemical gradients
- ☐ electrical gradients only
- ☐ ion concentration gradients only
- ☐ active transport pumps

Chapter 8 Question 54

Part A

The force driving simple diffusion is _____, while the energy source for active transport is _____.

ANSWER:

- ☐ phosphorylated carrier proteins; ATP
- ☐ transmembrane pumps; an electrochemical gradient
- ☐ a concentration gradient; ADP
- ☐ a concentration gradient; ATP hydrolysis

Chapter 8 Question 48

Part A

Which of the following molecules would decrease the electrochemical gradient across a membrane?

ANSWER:

- ☐ an aquaporin
- ☐ a sucrose carrier protein
- ☐ a proton pump
- ☐ a potassium channel

Chapter 8 Question 53

Part A

Several epidemic microbial diseases of earlier centuries incurred high death rates because they resulted in severe dehydration due to diarrhea. Today these diseases are usually not fatal because of the development of which of the following types of treatments?

ANSWER:

- ☐ intravenous feeding techniques
- ☐ antiviral medications that are efficient and work well with most viruses
- ☐ hydrating drinks with high concentrations of salt and glucose
- ☐ medications to slow blood loss

Chapter 8 Question 57

Part A

Which of the following processes is used by white blood cells when engulfing bacteria?

ANSWER:

- ☐ osmosis
- ☐ receptor-mediated exocytosis
- ☐ pinocytosis
- ☐ phagocytosis

Chapter 8 Question 46

Part A

The voltage across a membrane is called the _____.

ANSWER:

- ☐ membrane potential
- ☐ osmotic potential
- ☐ chemical gradient
- ☐ electrochemical gradient

Chapter 8 Question 7

EVOLUTION CONNECTION

Paramecium and other protists that live in hypotonic environments have cell membranes that limit water uptake, while those living in isotonic environments have membranes that are more permeable to water.

Part A

What water regulation adaptations might have evolved in protists in hypertonic habitats such as the Great Salt Lake?

ANSWER:

- ☐ Protists would have more permeable membranes than those in hypotonic or isotonic environments, allowing less water to enter the cell. Also, they could possibly develop lower salt concentration within the cell to retain water.
- ☐ Protists would have more permeable membranes than those in hypotonic or isotonic environments, allowing more water to enter the cell. Also, they could possibly develop higher salt concentration within the cell to retain water.
- ☐ Protists would have less permeable membranes than those in hypotonic or isotonic environments, allowing less water to enter the cell. Also, they could possibly develop higher salt concentration within the cell to retain water.
- ☐ Protists would have more permeable membranes than those in hypotonic or isotonic environments, allowing more water to enter the cell. Also, they could possibly develop lower salt concentration within the cell to prevent retention of water.

Part B

What water regulation adaptations might have evolved in protists in habitats with changing salt concentration?

Match the terms in the left column to the appropriate blanks in the sentences on the right. Terms can be used once, more than once, or not at all.

ANSWER:

Reset

Help

digestive vacuole

contractile vacuole

lysosome

not permeable

semi-permeable

water

not keep up

keep up

salt

Protists in a changing salt concentration will have a [] membrane in order to

[] with fluctuations in concentration and take in [] as they need, along with a

[] in order to pump out any unnecessary [] they take in.

Chapter 8 Question 39

Part A

Which of the following results is most likely result if a portion of an animal's blood is replaced with distilled water?

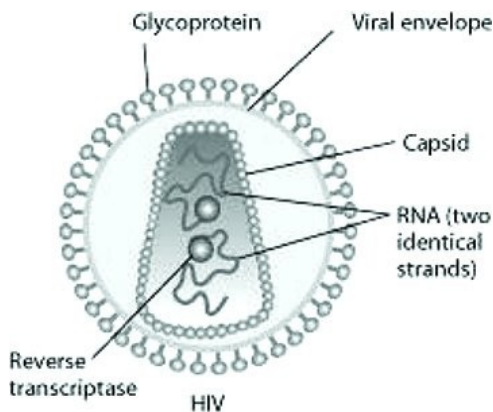
ANSWER:

- ☐ The animal's red blood cells will swell and possibly burst because the blood has become hypotonic compared to the cells.
- ☐ The animal's red blood cells will shrivel up because the blood has become hypotonic compared to the cells.
- ☐ The animal's red blood cells will shrivel up because the blood has become hypertonic compared to the cells.
- ☐ The animal's red blood cells will burst because the blood has become hypertonic compared to the cells.

Chapter 8 Question 62

Part A

Human immunodeficiency virus (HIV) infects cells that have both CD4 and CCR5 cell surface molecules. The viral nucleic acid molecules are enclosed in a protein capsid, and the protein capsid is itself contained inside an envelope consisting of a lipid bilayer membrane and viral glycoproteins. One hypothesis for viral entry into cells is that binding of HIV membrane glycoproteins to CD4 and CCR5 initiates fusion of the HIV membrane with the plasma membrane, releasing the viral capsid into the cytoplasm. An alternative hypothesis is that HIV gains entry into the cell via receptor-mediated endocytosis, and membrane fusion occurs in the endocytotic vesicle. To test these alternative hypotheses for HIV entry, researchers labeled the lipids on the HIV membrane with a red fluorescent dye.



What would be observed by live-cell fluorescence microscopy immediately after HIV entry if HIV enters the cell by endocytosis first, and then later fuses with the endocytotic vesicle membrane?

ANSWER:

- ☐ A spot of red fluorescence will diffuse in the infected cell's cytoplasm.
- ☐ The red fluorescent dye-labeled lipids will appear in the infected cell's interior.
- ☐ A spot of red fluorescence will remain outside the cell after delivering the viral capsid.
- ☐ A spot of red fluorescence will be visible on the infected cell's plasma membrane, marking the site of membrane fusion and HIV entry.

Chapter 8 Question 60

Part A

In receptor-mediated endocytosis, membrane receptors bind to molecules outside of the cell. Immediately following endocytosis where will the membrane receptors be located?

ANSWER:

- ☐ on the inside surface of the vesicle
- ☐ on the outer surface of the nucleus
- ☐ on the inside surface of the cell membrane
- ☐ on the outside of vesicles

Chapter 8 Question 59

Part A

Which of the following statements best describes the difference between pinocytosis and receptor-mediated endocytosis?

ANSWER:

- ☐ pinocytosis brings only water molecules into the cell, but receptor-mediated endocytosis brings in other molecules as well
- ☐ pinocytosis can concentrate substances from the extracellular fluid, but receptor-mediated endocytosis cannot
- ☐ pinocytosis increases the surface area of the plasma membrane, whereas receptor-mediated endocytosis decreases the plasma membrane surface area
- ☐ pinocytosis is nonselective in the molecules it brings into the cell, whereas receptor-mediated endocytosis offers more selectivity

Chapter 8 Question 56

Part A

An organism with a cell wall would most likely be unable to take in materials through which of the following processes?

ANSWER:

- ☐ facilitated diffusion
- ☐ phagocytosis
- ☐ active transport
- ☐ osmosis

Chapter 8 Question 61

Part A

Which of the following organelles plays a primary role in the digestion of bacteria engulfed by white blood cells?

ANSWER:

- ☐ vacuoles
- ☐ secretory vesicles
- ☐ Golgi vesicles
- ☐ lysosomes

Chapter 8 Question 58

Part A

Which of the following factors is a primary contributor underlying cause of familial hypercholesterolemia?

ANSWER:

- ☐ inhibition of the cholesterol active transport system in red blood cells
- ☐ defective LDL receptors on the cell membranes
- ☐ poor attachment of the cholesterol to the extracellular matrix of cells
- ☐ a poorly formed lipid bilayer that cannot incorporate cholesterol into cell membranes

Score Summary:

Your score on this assignment is 0.0%.

You received 0 out of a possible total of 76 points.